Activity: DAGs to Joint Probabilities

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Anant Deshwal[2]

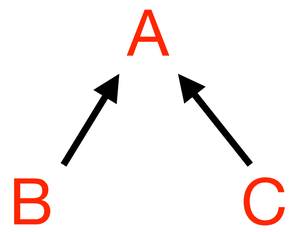
[1]University of Jyväskylä
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Credit to:

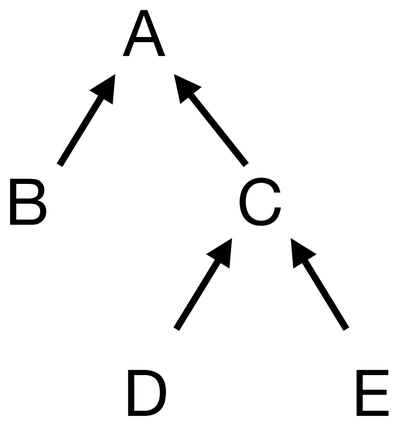
- Tom Hobbs (Colorado State University)
- Mevin Hooten (University of Texas)
- Becky Tang (Middlebury College)
- Alison Ketz (Funga)
- Slides adopted or based on the 2024 Short course on Bayesian Modeling for Ecologists, funded by NSF (*Award DEB 2042028 to Colorado State University*)

Convert following DAGs to Joint Distributions

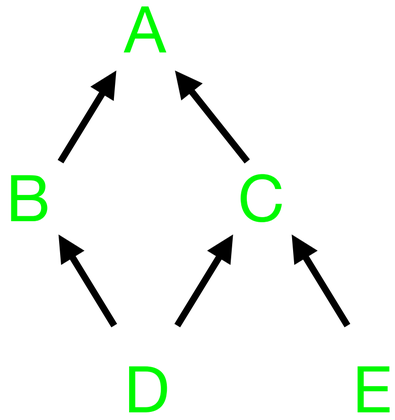
1)



2)

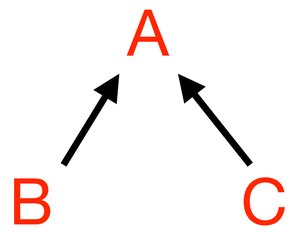


3)

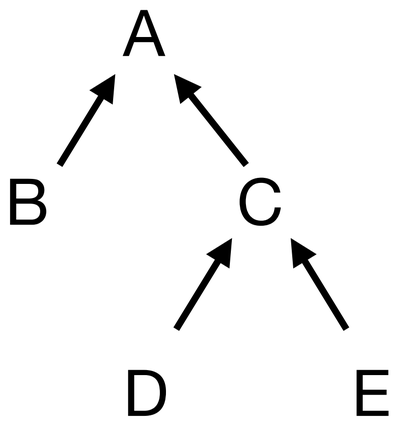


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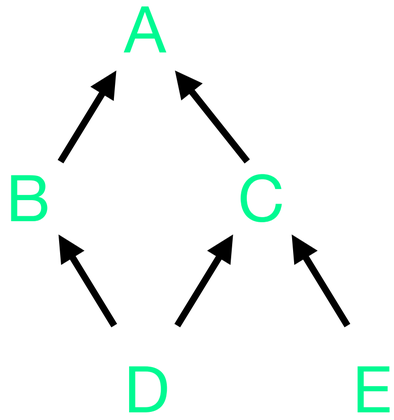
1)



2)



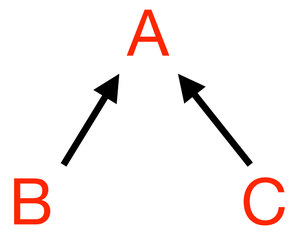
3)



[A,B,C] = [A|B,C] [B] [C]

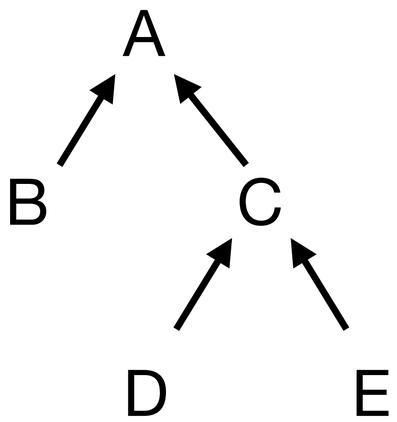
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1)

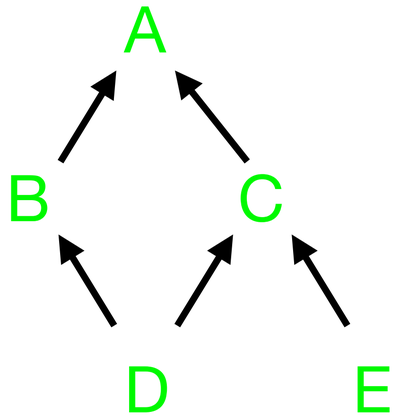


[A,B,C] = [A|B,C] [B][C]

2)



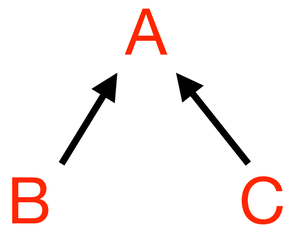
3)



[A,B,C,D,E] = [A|B,C] [C|D,E] [B] [D] [E]

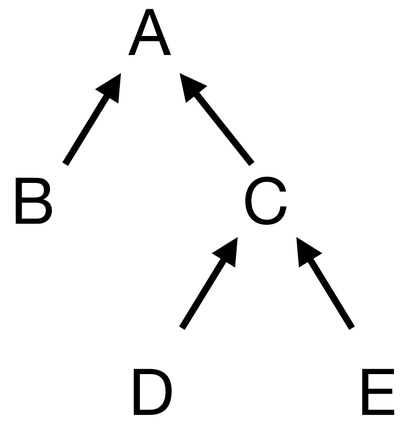
Convert following DAGs to Joint Distributions

1)



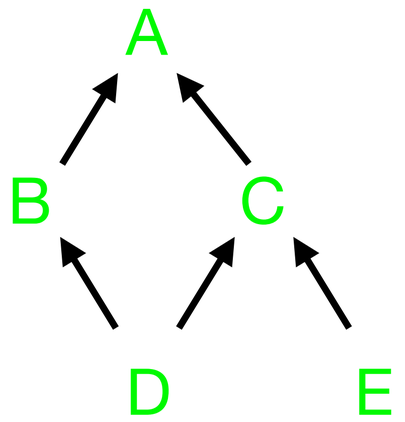
[A,B,C] = [A|B,C] [B][C]

2)



[A,B,C,D,E] = [A|B,C][C|D,E] [B][D][E]

3)



[A,B,C,D,E] = [A|B,C] [C|D,E] [B|D [D] [E]

Ecological Example

$$\frac{1}{N}\frac{dN}{dt} = r - \frac{r}{K}N,$$

- Per-capita rate of population Growth
  - r is intrinsic growth rate
  - K is the Carrying capacity. At N=K dN/dt =0
  - r/K is the strength of feedback from population size to population growth rate

Ecological Example

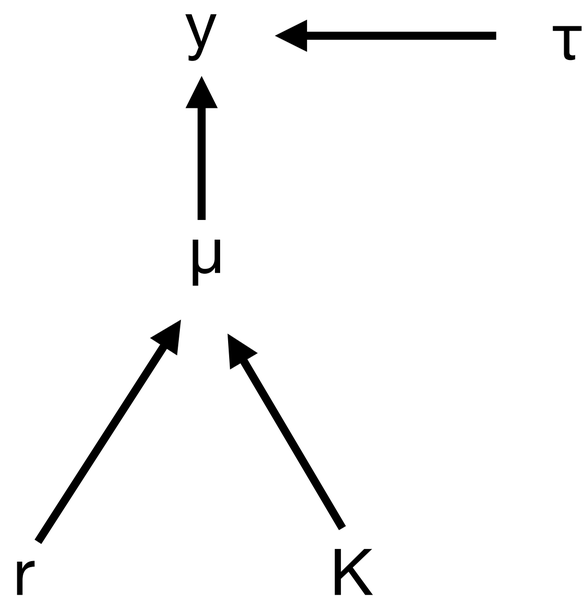
- Per-capita rate of population Growth
  - $y = 1/N_i \, dN_i/dt$
  - $x_i = N_i$
  - $\mu_i = r - rx_i/K$; where μ_i is the mean for y
  - What distribution would y take?

$$\frac{1}{N}\frac{dN}{dt} = r - \frac{r}{K}N,$$

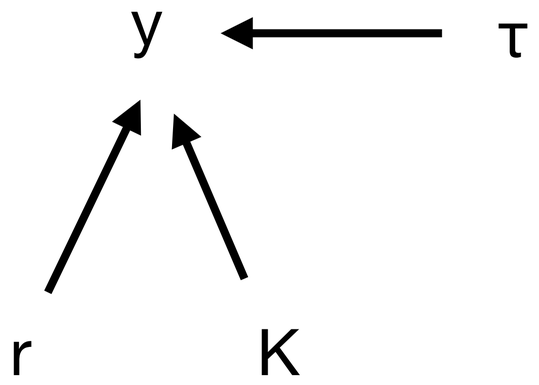
Ecological Example

- Per-capita rate of population Growth
  - $y = 1/N_i \, dN_i/dt$
  - $x_i = N_i$
  - $\mu_i = r - rx_i/K$; where μ_i is the mean for y
  - What distribution would y take - Normal
  - Draw DAG for it

$$\frac{1}{N}\frac{dN}{dt} = r - \frac{r}{K}N,$$



OR



$$[y,r,K] = [y|r,K]\,[r]\,[K]$$

- $[r, K, \tau|y] \propto \Pi_{i=1}^{n}$ normal $(y_i \mid \mu_i, \tau)$ gamma $(K| 0.001, 0.001)$ gamma $(\tau| 0.001, 0.001)$ gamma $(r|0.001, 0.001)$ … eqn 1
- $\mu_i = f(r, K, x_i) = r - rx_i/K$ …eqn2
- Eqn 1 can also be written as:
- $[r, K, \tau|y] \propto \Pi_{i=1}^{n}$ normal $(y_i \mid f(r, K, x_i), \tau)$ gamma $(K| 0.001, 0.001)$ gamma $(\tau| 0.001, 0.001)$ gamma $(r|0.001, 0.001)$
- Eqn 1 over here is the basis for your JAGs model
  - We are using τ (precision) instead of σ because JAGs uses τ
  - $\tau = 1/\sigma^2$

- $[r, K, \tau|y] \propto \Pi_{i=1}^{n}$ normal $(y_i \mid \mu_i, \tau)$ gamma $(K| 0.001, 0.001)$ gamma $(\tau|$ $0.001, 0.001)$ gamma $(r|0.001, 0.001)$ … eqn 1

- Structure: probability distribution (RV | parameters)

Per-Capita Growth Rate equation is Linear Regression

- $\mu_i = r - rx_i/K$
  - Is a Linear Regression equation $y = mx+b$,
    - $b = r$ and $m = -r/K$

JAGs Model

```
 ## Growth Model
Model {
        #Priors
        K ~ dgamma (0.001, 0.001)
        r ~ dgamma (0.001, 0.001)
        sigma ~ dunif (0, 2)
        tau <- 1/sigma^2 # Calculating precision from SD as JAGs understands Tau

        #Likelihood
        for(i in 1:n){
        mu[i] <- r- r/K* x[i]
        y[i] ~ dnorm(mu[i], tau)

        }
    }
```

$[r, K, \tau|y] \propto \Pi_{i=1}^{n}$ normal $(y_i \mid \mu_i, \tau)$ gamma $(K| 0.001, 0.001)$ gamma $(\tau| 0.001, 0.001)$ gamma $(r|0.001, 0.001)$ … eqn 1

$\mu_i = f(r, K, x_i) = r - rx_i/K$ …eqn2

- Write your JAGs model in a separate R script and save it eg: LogisticsJAGS. R

- You will use “LogisticsJAGS. R” in a separate R code to call to this JAGs model

Structure for JAGs Model

- Code for priors
- Code for deterministic model
- Code for likelihoods
- Priors and liklihoods are specified using ~
- We chose vague gamma priors for r, K, τ becuase they must be non-negative
- Loop replaces $\Pi_{i=1}^{n}$
-

Let’s Practice

Research question: How do temperature and precipitation influence plant height in grassland ecosystems?

Imagine you are an ecologist studying plant growth across 100 grassland plots. For each plot, you have measured:

Plant height (cm): Average plant height within each plot.

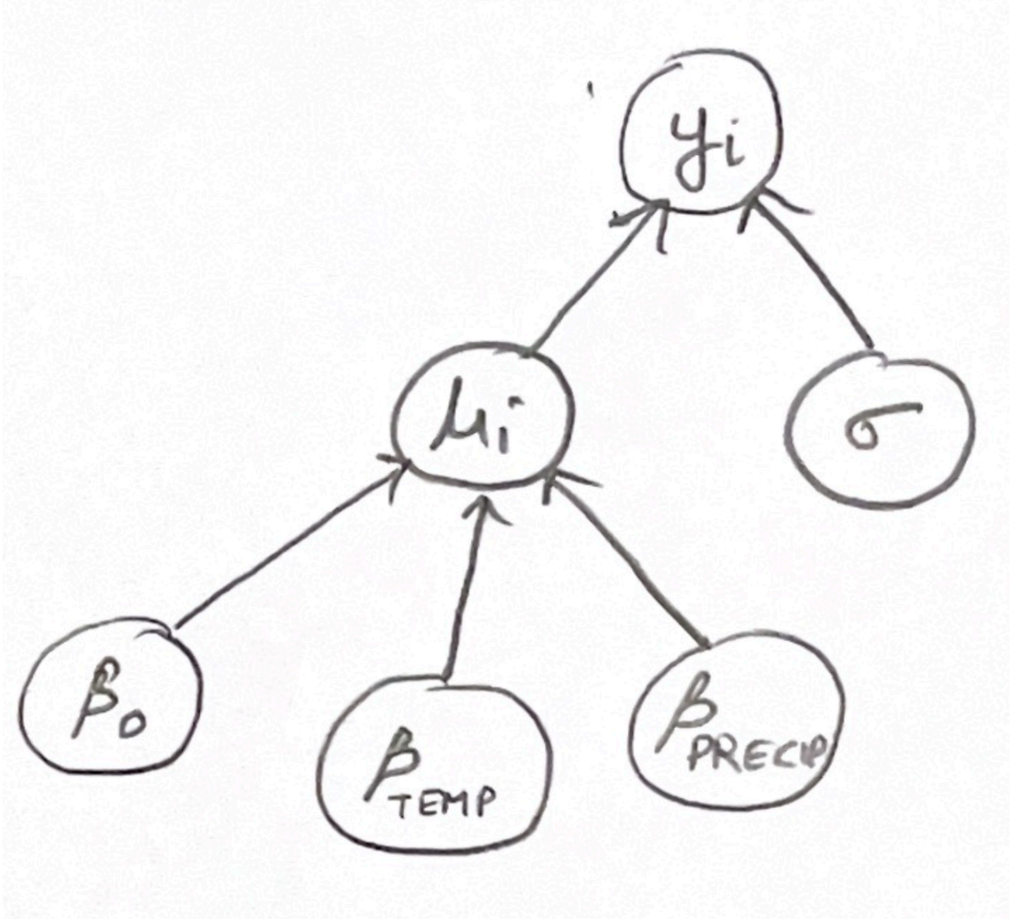
Temperature (°C): Mean growing-season temperature.

Precipitation (mm): Total growing-season precipitation.

You want to understand how temperature and precipitation influence plant height using a Bayesian linear regression model.

In-class Activity

- $y_i \sim \text{Normal}(\mu_i, \sigma^2)$
- $\mu_i = \beta_0 + \beta_1 T_i + \beta_2 P_i$
- $[\beta_0, \beta_1, \beta_2|y] \propto \Pi_{i=1}^{n}$ normal $(y_i | \mu_i, \tau)$ normal $(\beta_0| 0, 0.001)$ normal $(\beta_1| 0, 0.001)$ normal $(\beta_2|0, 0.001)$



DAG

JAGS code

```
model {
  # 1. PRIORS
  beta0 ~ dnorm(0, 0.0025)
  beta_temp ~ dnorm(0, 0.04)
  beta_precip ~ dnorm(0, 100)

  sigma ~ dunif(0, 20)
  # Convert standard deviation to precision
  tau <- 1 / (sigma * sigma)

  # 2. LIKELIHOOD
  for (i in 1:N) {
    plant_height[i] ~ dnorm(mu[i], tau)
    mu[i] <- beta0 +
             beta_temp * temperature[i] +
             beta_precip * rainfall[i]
  }
}
```